

Tippana Mani Deep Reddy

B.Tech - (2022-26)

Computer Science Engineering (AI&ML)

Swami Vivekananda Institute of Technology, Secunderabad

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PROFESSIONAL SUMMARY

Aspiring Backend Developer specializing in **Python** and **FastAPI**, with experience building scalable RESTful APIs during a successful internship at Codevocado . Proficient in **PostgreSQL**, **Docker**, and **Git**, with a strong focus on clean, efficient code and backend performance optimization. Passionate about applying AI/ML to solve complex data challenges in modern software ecosystems.

EDUCATION

Degree/Certificate	Institute/Board	CGPA/%	Year
B.Tech. CSE	Swami Vivekananda Institute of Technology, Secunderabad	8.20	2022–26
Senior Secondary	TSBIE Board	95.7%	2020–22
Secondary	SSC Board	9.8	2020

PROFESSIONAL EXPERIENCE

Web Developer Intern, Codevocado	<i>Sept. 2025 – Dec. 2025</i>
<i>Enterprise Resource Planning (ERP) System</i>	
- Engineered a scalable Backend Architecture for an Enterprise Resource Planning (ERP) system using Python and Flask, implementing JWT authentication and secure RESTful endpoints. - Automated PDF certificate and invoice generation workflows, reducing manual processing time by 70% and increasing operational throughput by 40%. - Participated in regular code reviews,debugging sessions and followed strict coding standards to ensure the delivery of clean, maintainable, and scalable code for 50+ users. - Tech Stack: Python, Flask, MySQL, JWT, REST APIs, Git.	

PROJECTS

Mutate AI	<i>Mar. 2026 - Present</i>
<i>Conversational AI Chatbot</i>	
- Engineered a chatbot using Python, FastAPI, and LangGraph featuring a complex state management system for seamless conversation exchange. - Implemented a dual-memory architecture: utilizing ChromaDB with cosine similarity embeddings for long-term semantic memory and PostgreSQL for reliable short-term session storage. - Optimized information retrieval through vector-based search, significantly improving the relevance and accuracy of chatbot responses in multi-turn dialogues.	
Calorie Burn Prediction System	<i>Nov. 2025 – Feb. 2026</i>
<i>AI-based Fitness Tracking Web Application</i> Github	
- Developed a real-time calorie burn prediction engine utilizing computer vision, effectively replacing wearable fitness trackers and reducing user costs by up to 60%. - Implemented Role-Based Access Control (RBAC) for Admin and User management, ensuring secure data isolation and personalized insights. - Optimized database schemas in MySQL to support efficient storage and retrieval of fitness metrics, reducing latency by 30%.	
Question Papers Hub	<i>Feb. 2025 – Mar. 2025</i>
<i>Generative AI Document Management System</i> Github	
- Engineered a Generative AI backend using Python to analyze PDF documents, utilizing Natural Language Processing (NLP) to identify complex patterns and extract key insights. - Built robust REST APIs to facilitate seamless communication between the AI engine and the frontend, reducing document research time by 90%.	

TECHNICAL SKILLS

Backend Frameworks: Python, Flask, FastAPI, RESTful APIs, Object-Oriented Programming (OOP), Asynchronous Programming (Asyncio), SQLAlchemy (ORM) AI & LLM Orchestration: LangGraph, OpenAI API, NLP, Vector Databases (ChromaDB), Embeddings Databases: PostgreSQL, MySQL, Database Design and Optimization DevOps Tools: Docker, Git/GitHub, Nginx/Gunicorn Methodologies Testing: Agile, Scrum, System Design, Code Reviews Frontend Technologies: React.js, JavaScript (ES6+), Tailwind CSS

CERTIFICATIONS

Udemy: Entry-Level Python Programmer, Associate in Python Programming Galaxy Institute of Information Technology: Diploma in Computer Application and Programming(D-CAP)
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